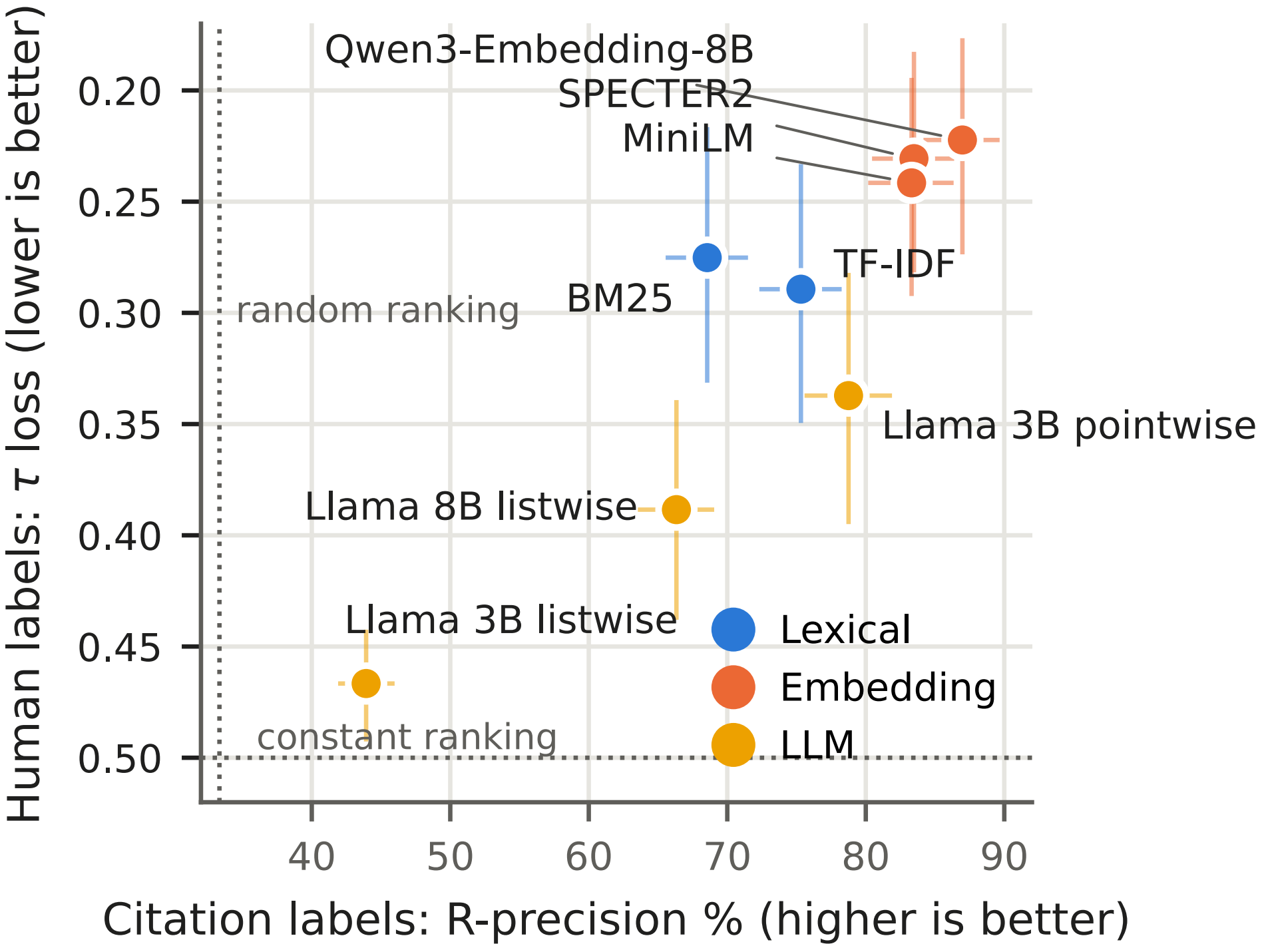
Qwen3-Embedding-8B
SPECTER2
MiniLM
TF-IDF
BM25
random ranking
Llama 3B pointwise
Llama 8B listwise
Llama 3B listwise
constant ranking
Lexical
Embedding
LLM
0.20
0.25
0.30
0.35
0.40
0.45
0.50
40
50
60
70
80
90
Human labels: τ loss (lower is better)
Citation labels: R-precision % (higher is better)